

BACKGROUND PLACEHOLDER
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Quality Control (QC) Audit Report

Software Requirements Specification (SRS)

Abyss Navigator Suite Project

Veridion Systems Validation Labs

March 17, 2026

Brigman Team Member	Department	Role	Employee ID
Muhammad Ridwan Putra Jasni	QC Team	QC Audit Lead	QC-001
Lin Yuhao	QC Team	QC Specialist	QC-002
Muhammad Mikhail Bin Mazlan	QC Team	QC Specialist	QC-003
Kannan s/o Rajamohan	QC Team	QC Specialist	QC-004

Table 1: QC Team – Veridion Systems Validation Labs

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1 Executive Summary

This report presents the results of a Quality Control (QC) validation of the Software Requirements Specification (SRS) for the **Abyss Navigator Suite**. The objective of this QC audit is to identify defects that remain in the SRS after the completion of Eliciting, Analysing, and Specifying.

The audit focused on three principal categories of SRS defects expected at the Validating stage:

- **Authorship defects** – where intended meanings are not expressed clearly enough
- **Omission defects** – where required scenarios, conditions, or supporting content are missing
- **Contradiction defects** – where different parts of the SRS imply inconsistent behaviour

The inspection found that the SRS is generally well-structured and substantively mature. However, a number of issues were detected in wording precision, edge-case coverage, control-behaviour consistency, and traceability coherence.

These findings indicate that the SRS is **conditionally acceptable**, provided that the identified defects are rectified before the document is baselined for downstream design and implementation.

2 Introduction

2.1 Purpose

This Quality Control (QC) Audit Report presents the findings of an independent inspection of the Software Requirements Specification (SRS) prepared for the **Abyss Navigator Suite**. The purpose of this QC activity is to identify defects remaining in the SRS work product before the project progresses into downstream stages such as system design and implementation.

In accordance with the validating stage of Requirements Engineering, QC focuses on defect detection in the **requirements work product itself**. The QC team records findings and recommended rectifications but does not directly modify the SRS.

2.2 Document Conventions

This document is an audit report rather than a replacement SRS. It records detected defects, explains their implications, and recommends rectification actions.

Audit findings use the format QC-FND-XXX.

Rectification recommendations use the format QC-REC-XXX.

Error detection instances use the format QC-ERR-XXX.

2.3 Intended Audience and Reading Suggestions

This report is intended for the Brigman requirements engineering team, project management personnel, and relevant Triton Deepwater stakeholders involved in SRS review and approval.

Readers should begin with the Executive Summary and Sections 1–3, then proceed to the QC Checklist, Error Detection, detailed Findings, and Rectification Recommendations.

2.4 Audit Scope

The audit covers the SRS as a QC exercise. The scope includes:

- detection of indiscriminate authorship defects
- detection of latent contradictions
- detection of omission of topics
- checking codification and traceability coherence
- checking structural, editorial, and presentation consistency

3 Audit Basis

3.1 Basis of Evaluation

This QC audit was conducted on the understanding that the SRS should already be in reasonably strong condition after completion of Eliciting, Analysing, and Specifying. The purpose of QC at this stage is therefore not to rewrite the SRS, but to detect remaining defects still likely to be present in the work product.

3.2 Nature of QC

Quality Control focuses on inspecting the SRS work product itself. The QC team identifies defects but does not directly amend the SRS. Instead, the audit team produces a report that identifies defects and recommends rectifications for implementation by the requirements engineering team.

3.3 Kinds of Errors Pursued

The QC audit focuses on detecting the following major defect categories expected in the SRS at the validating stage:

- **Indiscriminate side effects of authorship** – where intended meaning appears clear to the original writers but is not expressed clearly enough in the written requirement
- **Latent contradictions** – where requirements, models, or sections of the SRS may conflict with one another
- **Omission of topics** – where expected requirements content, edge conditions, or supporting information is absent
- **Traceability coherence issues** – where codification, source mapping, or cross-references are incomplete or unclear
- **Editorial and structural inconsistencies** – where formatting, section naming, or presentation patterns weaken readability or professionalism

3.4 Severity Classification

Findings are categorised according to severity:

- **High** – defect that may significantly affect interpretation, implementation, or expected system behaviour

- **Medium** – defect that may introduce ambiguity or reduce clarity without fundamentally changing requirement intent
- **Low** – minor editorial or structural issue that does not materially affect requirement meaning

3.5 Documents Reviewed

The audit considered the following project materials as supporting references:

- Software Requirements Specification (SRS)
- Project Specification (PS)
- relevant requirements models
- codification and iRTM sections
- quality model and change mitigation sections
- appendices and supporting traceability artefacts

4 QC Plan

4.1 Target Workproduct

The target workproduct for this QC audit is the released **Software Requirements Specification (SRS)** for the Abyss Navigator Suite project.

4.2 Context and Goals

The SRS is the main requirements work product produced after Analysing and Specifying. Since downstream design and implementation rely on the SRS as the controlling requirements baseline, the QC audit is conducted to provide confidence that the specification is sufficiently clear, complete, and internally consistent.

The goals of this QC audit are:

- to determine whether the SRS clearly and consistently represents the intended requirements
- to identify material defects before downstream design and implementation begin
- to recommend rectifications that improve correctness, completeness, coherence, and readability of the SRS

4.3 Access and Audit Conditions

The QC team reviewed the released SRS and supporting project artefacts made available for audit. Access was limited to materials needed to inspect the work product and its supporting traceability and modelling references.

4.4 Inspection Techniques Used

The following investigative techniques were applied during the QC audit:

1. **Close document inspection** – detailed reading of requirement clauses and supporting sections

2. **Perspective-based reading** – reviewing the SRS from likely reader viewpoints such as developer, reviewer, and stakeholder
3. **Cross-section contradiction checking** – comparing different SRS sections and models to detect hidden conflicts
4. **Codification and traceability verification** – checking requirement IDs, source links, and mapping consistency
5. **Model-to-text consistency inspection** – ensuring models and textual requirements express compatible meanings
6. **Omission mapping against expected SRS topics** – checking whether important topics, edge cases, and exception paths are sufficiently covered

4.5 Work Breakdown

Role	Assigned Member	Primary Responsibility
QC Audit Lead	Muhammad Ridwan Putra Jasni	Coordinate audit activities, consolidate findings, and ensure consistency of final report writing.
Requirements Reviewer	Lin Yuhao	Inspect operational, technology, and quality-related sections for ambiguity, contradiction, and omission.
Traceability Reviewer	Muhammad Mikhail Bin Mazlan	Review codification, iRTM references, source mappings, and section-to-section coherence.
Editorial and Structure Reviewer	Kannan s/o Rajamohan	Review document structure, section naming, formatting consistency, and overall presentation quality.

Table 2: QC audit team work allocation

4.6 Indicative Timeline

Stage	Activity	Indicative Duration
Stage 1	Individual inspection of assigned SRS sections and supporting artefacts	2–3 hours
Stage 2	Consolidation meeting and de-duplication of identified findings	1 hour
Stage 3	Severity review and drafting of rectification recommendations	1 hour
Stage 4	Final report formatting, checking, and team sign-off	1 hour

Table 3: Indicative QC audit timeline

4.7 Reporting

The QC team records findings and rectification recommendations in this report. The audit team does not directly amend the audited SRS.

5 QC Checklist

The following checklist was used by the QC audit team to inspect the SRS work product. The checklist items represent critical aspects of a high-quality SRS.

Item	Checklist Question	Inspection Result
C1	Are all major SRS sections present and logically structured?	Mostly satisfied
C2	Are requirements written with clear trigger conditions and expected system responses?	Partially satisfied
C3	Are normal, alternate, and exception scenarios sufficiently covered?	Partially satisfied
C4	Are requirement IDs consistently codified and traceable to source clauses and models?	Mostly satisfied
C5	Are requirement statements measurable and testable?	Partially satisfied
C6	Are system behaviours consistent across textual descriptions and supporting models?	Partially satisfied
C7	Are terminology, naming conventions, and formatting patterns used consistently?	Mostly satisfied
C8	Are addenda and supporting artefacts appropriately linked to the base report?	Mostly satisfied

Table 4: QC inspection checklist

6 Error Detection

The QC audit identified instances of three principal QC error types: authorship, omission, and contradiction. Each detected instance is recorded with its location and characteristic.

Error ID	Error Type	Location	Characteristic of Error
QC-ERR-001	Contradiction	Section 3.1 UR-MOB-06 vs Section 4.2 User Authentication vs Section 5.3 NFR-SEC-03	The UI requirements say parents and teachers authentication shall include 2FA; however, the authentication feature in Section 4.2 only specifies username/password plus session token for parents/teachers, while Section 5.3 explicitly requires 2FA only for administrators. This creates conflicting statements on who actually must use 2FA.
QC-ERR-002	Contradiction	Section 4.5 Attendance Notification	The description and stimulus/response for this feature talk only about notifications to parents, but FR-NOTIF-09 later adds teacher push notifications for attendance events. That broadens the feature scope in a way not reflected in the section description or sequence.

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Error ID	Error Type	Location	Characteristic of Error
QC-ERR-003	Omission	Section 2.7 Assumptions and Dependencies	The SRS says data mapping specifications are a pending open issue, but Appendix B's official open issues list contains only CM-OI-01 to CM-OI-03 and does not include this data-mapping issue. So an unresolved issue is mentioned but not formally tracked.
QC-ERR-004	Omission	Section 4.2 User Authentication	The document states that accounts lock after failed login attempts, but there is no clear functional requirement for password reset / recovery flow for parents and teachers after lockout. Since authentication and account lockout are specified, recovery handling is a notable missing topic.
QC-ERR-005	Authorship	Section 1.3 Intended Audience and Reading Suggestions	The Project Manager is told to read Section 6 for the iRTM and change mitigation strategy, but those topics are actually in Section 7 and Section 8. This is a cross-reference error caused by inconsistent document wording/editing.
QC-ERR-006	Contradiction	Section 4.4 Leave Submission vs FR-LEAVE-03	FR-LEAVE-02 says a leave request shall not be recorded if MCS validation fails or is unresolved, but FR-LEAVE-03 says the system shall record all leave submissions with timestamp, user ID, student ID, reason, and MCS outcome. That can be read as conflicting: is a failed submission recorded as a leave request, or not recorded at all?
QC-ERR-007	Omission	Section 5.2 Safety / Section 6.1 Database / Section 8.2 Open Issues	Disaster recovery is acknowledged as dependent on unresolved RPO/RTO, but the SRS does not clearly state the actual recovery behavior, restoration sequence, or failover handling while the issue remains open. Given the importance of attendance data, the recovery topic is under-specified.
QC-ERR-008	Authorship	Section 5.4.1 QA-USAB-01	The clause says "The system shall allow parent and teacher users must be able to...", which is grammatically malformed. This is a classic authorship error where intended meaning is clear, but the exact text is poorly expressed.

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Error ID	Error Type	Location	Characteristic of Error
QC-ERR-009	Omission	Section 4.3 Attendance Information Access / Section 4.6 Oversight	Attendance viewing and reporting are covered, but there is no clearly defined requirement for correction workflow when an attendance record is wrong other than the scan-failure manual override. A fuller exception-handling path for disputed records appears missing.
QC-ERR-010	Authorship	Section 3.1 UR-WEB-05	The requirement is compound / incongruent because it combines multiple distinct functions in one clause: a reports screen, several report categories, and CSV/PDF export behaviour. This reduces atomicity, clarity, and testability.
QC-ERR-011	Authorship / Contradiction	Section 3.1 UR-WEB-06 vs Section 4.1 FR-RBAC-04	The requirement is compound / incongruent because it combines listing users with add, edit, and delete operations in one clause. It is also potentially inconsistent with FR-RBAC-04, which specifies account deactivation and retention for audit, making the wording "delete users" questionable.
QC-ERR-012	Authorship	Section 3.3 FR-SW-02	The requirement is compound / incongruent because it bundles timeout policy, retry behaviour, failure logging, and administrator notification into one clause. These are distinct technical behaviours expressed together rather than as atomic requirements.
QC-ERR-013	Authorship	Section 4.1 FR-RBAC-03	The requirement is compound / incongruent because it groups role assignment updates, credential resets, account unlocks, and change logging into one clause. These are separate administrative behaviours and reduce atomicity when kept together.
QC-ERR-014	Authorship	Section 4.6 FR-OVS-02	The requirement is mildly compound / incongruent because it combines threshold configuration, immediate effect of changes, and audit logging in one clause. The meaning is understandable, but it is less atomic than the surrounding requirements.

Table 5: Detected QC error instances

7 QC Findings

7.1 Finding Summary

Finding ID	Type	Severity	Short Description
QC-FND-001	Authorship defect	Medium	Some requirement statements remain broad enough that different technically competent readers may interpret them differently.
QC-FND-002	Latent contradiction	High	Certain statements imply autonomous behaviour by default while other statements imply supervisory or intervention behaviour that may conflict with that autonomy.
QC-FND-003	Omission of topic	High	Exceptional scenarios, boundary conditions, and failure-state handling are not always specified to the same depth as normal operations.
QC-FND-004	Traceability coherence	Medium	Some links between requirements, models, and traceability references would benefit from tighter one-to-one clarity.
QC-FND-005	Editorial / structural consistency	Low	Minor inconsistencies in naming, formatting, and section presentation reduce document professionalism and readability.

Table 6: Summary of QC findings

7.2 Detailed Findings

QC-FND-001: Ambiguity caused by authorship side effects

Related error detection: QC-ERR-001

Type: Authorship defect

Severity: Medium

Context. The relevant requirement text is intended to support remote intervention behaviour under constrained communications.

Result. The wording remains broad enough that two competent readers may derive slightly different implementation expectations.

Analysis. This is characteristic of authorship-side defects in which intended meaning is clear to the writers but not expressed with sufficient precision in the written requirement.

Message. Rewrite the affected statements to improve measurability, precision, and trigger conditions without changing the original intended requirement meaning.

QC-FND-002: Possible contradiction between autonomous and supervised behaviour

Related error detection: QC-ERR-002

Type: Latent contradiction

Severity: High

Context. The SRS strongly emphasises autonomous operation as a core business and technical expectation.

Result. Other passages also imply human intervention or supervision behaviour that could be interpreted differently depending on operational circumstances.

Analysis. This creates a latent contradiction because the control precedence between autonomy and supervision is not yet expressed with sufficient singularity.

Message. Introduce a single, explicit control hierarchy defining default autonomy, intervention thresholds, takeover conditions, and precedence during abnormal or emergency states.

QC-FND-003: Incomplete treatment of exception and edge scenarios

Related error detection: QC-ERR-003

Type: Omission of topic

Severity: High

Context. Normal operating behaviour is generally specified more strongly than abnormal, degraded, or edge-case operation.

Result. Some failure and recovery conditions are not described to the same depth as ordinary operating behaviour.

Analysis. This omission creates a completeness risk because downstream design and implementation may fill the gaps using assumptions rather than agreed requirements.

Message. Add explicit clauses or elaborations covering exception paths, degraded states, recovery boundaries, timeout conditions, and data-loss contingencies.

QC-FND-004: Traceability and codification tightening needed

Related error detection: QC-ERR-004

Type: Traceability coherence

Severity: Medium

Context. The SRS includes codification and traceability mechanisms intended to support downstream auditability.

Result. Some areas would benefit from stronger one-to-one linking between requirements, models, and verification references.

Analysis. Auditability improves when each requirement is linked clearly to source clauses, supporting models, and traceability references.

Message. Review all requirement IDs, source mappings, use-case links, and iRTM references to ensure consistency, completeness, and unambiguous downstream tracing.

QC-FND-005: Minor presentation and naming inconsistencies

Related error detection: QC-ERR-005

Type: Editorial / structural consistency

Severity: Low

Context. The SRS is generally well structured and professionally presented.

Result. Some presentation details such as repeated heading styles, naming patterns, and table conventions can still be improved.

Analysis. While not materially changing requirement meaning, such inconsistencies weaken readability and professionalism.

Message. Standardise section naming, heading capitalisation, cross-reference style, and table formatting throughout the document.

8 Rectification Recommendations

8.1 Priority Rectification Actions

Based on the defects identified during the QC inspection, the following rectification actions are recommended in priority order:

1. resolve the autonomy-versus-intervention contradiction by introducing explicit control precedence
2. strengthen edge-case and exception-path coverage
3. revise ambiguous wording in measurable requirements
4. tighten codification and traceability mappings
5. standardise presentation and editorial consistency

8.2 Recommended Ownership of Corrections

Recommendation ID	Owner	Action
QC-REC-001	Requirements Engineering Lead	Coordinate revision of all high-severity findings and confirm alignment with customer intent.
QC-REC-002	Requirements Review Team	Reword ambiguous requirements and add omitted exception-handling clauses.
QC-REC-003	Traceability Reviewer	Recheck codification, source mapping, use-case links, and iRTM consistency.
QC-REC-004	Editorial Reviewer	Apply consistent formatting, naming, and reference conventions.

Table 7: Recommended ownership of rectification actions

9 Conclusion

The QC audit concludes that the Software Requirements Specification is substantively mature and reflects strong work carried out during the earlier Requirements Engineering stages. However, the document still contains defects typical of the validating stage, particularly in the areas of authorship side effects, latent contradictions, omission of edge topics, and traceability tightness.

The QC team does not directly modify the SRS. Instead, the findings and rectification recommendations documented in this report should be implemented by the requirements engineering team to strengthen document clarity, completeness, and internal coherence before the SRS is baselined for downstream development.

Overall, the SRS is assessed as **conditionally acceptable subject to rectification of the identified findings**.

10 Sign-off

10.1 QC Audit Team Authentication

Name	Role	Authentication
Muhammad Ridwan Putra Jasni	QC Audit Lead	Submitted by named team member
Lin Yuhao	QC Auditor	Submitted by named team member
Muhammad Mikhail Bin Mazlan	QC Auditor	Submitted by named team member
Kannan s/o Rajamohan	QC Auditor	Submitted by named team member

Table 8: QC audit team sign-off

11 Addenda

11.1 Repository Information

PROJECT REPOSITORY https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113 <i>QC report, SRS, models, and traceability artefacts maintained under version control</i>
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11.2 Glossary

Term	Definition
QC	Quality Control inspection of a work product, in this case the SRS.
SRS	Software Requirements Specification used as the requirements baseline for downstream development.
Authorship defect	Defect caused by unclear or imprecise expression of intended requirement meaning.
Omission defect	Defect in which required content, scenario, or supporting information is missing.
Contradiction defect	Defect in which different parts of the SRS imply inconsistent or conflicting behaviour.
Traceability coherence	The degree to which requirements, models, sources, and references link together clearly and consistently.

Table 9: QC glossary addendum

References

- Triton Deepwater, *Project Specification: The Abyss Navigator Suite*, ICT2113, 2026.
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- Karl E. Wiegers, *Software Requirements Specification Template*, 1999.